

Design for Inattention^{*}

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Abstract

We analyze a model in which a principal designs a risky option for a rationally inattentive agent who can choose a safe outside option. The principal controls payoffs across states to maximize acceptance probability, while the agent can acquire costly information about the state before deciding. Because the stakes determine the value of information, the principal can manipulate information acquisition through option design. We show that as the environment becomes more certain, an acceptance-motivated principal moderates risk to discourage learning. In contrast, a benevolent principal would always offer maximal risk to maximize the agent's option value. Most strikingly, the agent's welfare can be non-monotonic in environmental certainty, decreasing once predictability becomes sufficiently high.

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JEL classification codes: D82, D83, D86

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1 Introduction

When a principal designs a risky option for an agent who can pay to learn about its value, the stakes she offers determine not only the option’s attractiveness but also how much the agent learns. The principal controls the distribution of payoffs across states and seeks to maximize the probability that the agent accepts the risky option. After observing the payoff profile, a rationally inattentive agent chooses how much information to acquire about the state of the world and then decides whether to accept or instead choose a safe outside option. The value of information, therefore, depends not only on the underlying uncertainty but also on the stakes associated with different outcomes. When these stakes are chosen strategically, information acquisition and option design are jointly determined.

We characterize the principal’s optimal design and show that it exhibits risk moderation as the environment becomes more certain. When prior uncertainty about the state is high, the principal must offer a maximally risky option to compete with the safe outside alternative, thereby inducing information acquisition. As the prior probability of the favorable state increases, the principal can instead reduce payoff dispersion while keeping the risky option sufficiently attractive for acceptance. This moderation lowers the value of information and discourages learning. Greater ex-ante certainty, therefore, expands the scope for suppressing information acquisition through stake design.

This strategic response has sharp welfare implications. We show that a benevolent designer, one who maximizes the agent’s expected utility, would always choose maximal payoff dispersion, as doing so maximizes the option value of the risky alternative under rational inattention. By contrast, an acceptance-motivated principal moderates risk whenever feasible. As a result, the agent’s equilibrium welfare can be non-monotonic in the predictability of the state of the world, decreasing once predictability becomes high enough that moderation is optimal. Thus, greater predictability, which would ordinarily benefit a decision-maker, instead harms the agent by enabling designs that suppress information acquisition and lower expected payoff.

The remainder of the paper is organized as follows. In the next section, we review the related literature. Section 3 presents our model, which we then analyze in Section 4. Section 5 explores agent welfare. Section 6 concludes.

2 Related literature

This paper contributes to the literature on rational inattention initiated by Sims (2003), in which agents face costs of acquiring and processing information. A large body of work studies how such costs shape optimal decision-making and stochastic choice when payoffs are exogenously given (see, e.g., Matějka and McKay (2015); Caplin et al. (2022); Maćkowiak et al. (2023)). We focus on how the strategic design of payoffs affects the value of information.

The paper is most closely related to the literature on security design and contracting with endogenous information acquisition. Crémer and Khalil (1992) show that when precontractual information acquisition is purely wasteful, the principal always designs the contract to deter it.

Cr  mer et al. (1998) extend this to productive information gathering and show that whether the principal induces or deters learning depends on the cost of information. Szalay (2009) develops a broader framework in which more sensitive, higher-powered contracts encourage information acquisition, whereas less sensitive contracts discourage it. Yang and Zeng (2019) and Yang (2020) show that payoff structures affect incentives to acquire information and characterize optimal securities when information acquisition is flexible.¹ Our analysis differs in two fundamental respects. First, we study a setting in which the principal directly chooses the option’s state-contingent payoff profile, rather than selecting among technologically constrained securities or contracts, and is purely acceptance-motivated: information acquired by the agent has no productive or allocative value for the principal. Second, and most importantly, we isolate a welfare channel absent from this literature: greater ex-ante certainty can reduce the agent’s welfare because it enables the principal to shift from inducing to suppressing information acquisition, compressing payoff dispersion in ways that lower the agent’s option value.

Our results also relate to rational inattention in strategic environments where improvements in the information environment trigger offsetting strategic responses. In Trombetta (2020), voters can allocate costly attention between politicians’ actions (policies) and the underlying state of the world. Because politicians pander more when actions are more likely to be observed, cheaper attention can increase pandering and reduce voter welfare. In Martin (2017), an informed seller sets prices that may reveal product quality to a rationally inattentive buyer; when attention becomes cheaper (and the outside option is sufficiently attractive), low-quality sellers may mimic more, making prices less informative. These papers show that making information acquisition easier need not improve outcomes once other players respond strategically. We complement this insight by identifying a different channel: in our model, the relevant comparative static is not a reduction in information costs but an increase in ex-ante certainty about fundamentals. Greater certainty can reduce the agent’s welfare because it enables the principal to compress payoff dispersion and thereby discourage learning.

Another related strand of literature examines environments in which principals influence outcomes by controlling information structures or disclosure policies, often referred to as Bayesian persuasion with costly information acquisition (see, e.g., Gentzkow and Kamenica (2014); Bloedel and Segal (2021); Matyskova and Montes (2023); Lipnowski et al. (2022b)). In these models, the principal manipulates beliefs by choosing what information the agent receives or how it is revealed. By contrast, in our setting, the principal does not directly control information or beliefs. The agent remains free to acquire any information about the state, subject only to her information cost. Manipulation of attention arises instead through stake design: by reshaping the payoff profile of the risky option, the principal changes the marginal return to information acquisition.

Finally, on a technical level, the paper belongs to the class of rational inattention models with

¹See also Avoyan and Romagnoli (2023), who propose a novel method for eliciting the attention level solely by observing the decision-maker’s incentive redistribution choice. Similar to the mechanism in our paper, they show that by reducing the gap between payoffs in different states, the decision-maker, who can directly influence the payoff distribution across states, can affect their incentives to pay attention: the smaller the gap, the less attentive the decision-maker needs to be.

posterior-separable information costs. We use a quadratic cost for tractability, following Wei (2021); Lipnowski et al. (2022a) and Jain and Whitmeyer (2020)², and verify that all qualitative results extend to Shannon entropy costs³ (Shannon, 1948; Cover and Thomas, 2012).

3 The model

This section describes the game’s general setup, the agent’s problem, and the principal’s option design problem. We also introduce the timing of the game and the equilibrium.

3.1 Description of the setup

An agent faces a discrete choice between a safe (S) and a risky (R) option. There are two states of the world $\Omega = \{\omega_L, \omega_H\}$, with $\omega \in \Omega$ denoting a generic state. The safe option provides a known payoff $S > 0$ to the agent in both states of the world. Before the agent’s choice is made, the principal designs a risky option, i.e., the state-dependent utility the option delivers to the agent if accepted. The principal selects the option design to maximize the probability of acceptance. We fully specify the principal’s problem in subsection 3.3. The agent’s action $a \in A = \{S, R\}$ is to choose between the safe option and the risky option. We denote $v(a|\omega) : A \times \Omega \rightarrow \mathbb{R}_+ \cup \{0\}$ ⁴ as the payoff to the agent from selecting action $a \in A$ in state $\omega \in \Omega$. Hence $v(S|\omega) = S$, $\forall \omega \in \Omega$ and $v(R|\omega)$ is chosen by the principal for each state $\omega \in \Omega$.

The realization of the state of the world $\omega \in \Omega$ is unknown to both the principal and the agent. They share a common prior belief about the state realization. Let $\mu \in [0.5, 1)$ denote the probability of state ω_H , so $1 - \mu$ is the probability of state ω_L .⁵ For notational simplicity, we write μ rather than $\mu(\omega_H)$ throughout. The prior μ thus reflects the certainty about which state will be realized: $\mu = 0.5$ represents maximal uncertainty, while $\mu \rightarrow 1$ represents near-certainty that ω_H will occur.

We model the agent as rationally inattentive (Sims, 2003). Before making her decision, she can acquire a costly signal $x \in \mathbb{R}$. The joint distribution of the signals and states, $f(x, \omega) \in \Delta(\mathbb{R} \times \Omega)$, where $\Delta(\mathbb{R} \times \Omega)$ is the set of all probability distributions on $\mathbb{R} \times \Omega$, is subject to the agent’s choice. Reducing uncertainty to a greater extent is increasingly costly. After the agent receives a signal from the selected information structure, she updates her belief using Bayes’

²See also Ely et al. (2015); Augenblick and Rabin (2021) who use the quadratic difference between prior and posterior in utility functions.

³See Appendix H.

⁴Restricting payoffs to be non-negative is without loss of generality. The agent’s incentives to acquire information depend only on payoff differences across states, not on their absolute level. Any payoff structure with negative values can therefore be shifted by adding a constant to all state-contingent payoffs without affecting information acquisition or the value of information. Such a shift must be accompanied by a corresponding shift in the budget parameter. The payoffs non-negativity condition thus serves as a convenient normalization and can be interpreted as a limited-liability condition.

⁵We focus on $\mu \geq 0.5$ without loss of generality, as the model is symmetric around $\mu = 0.5$. The case $\mu < 0.5$ is analogous to the roles of ω_L and ω_H reversed.

rule and chooses an action $a \in A$. Her choice rule is modeled as $x \rightarrow \sigma(x) \in A$. The agent's objective is to maximize the expected payoff less the cost of information.

3.2 The agent's decision problem

Each signal realization is associated with a corresponding posterior belief about the state. Therefore, an information-processing strategy induces a distribution over posterior beliefs about the state of the world. As shown by Kamenica and Gentzkow (2011) and Caplin and Dean (2013), instead of specifying the information-processing selection problem, we can equivalently work with the distributions of the posterior beliefs that average out to the prior. Note that, under the optimal strategy, there is a one-to-one mapping between a signal and an action because obtaining multiple signals leading to the same action is wasteful and thus sub-optimal (Matějka and McKay, 2015). Consequently, instead of working with signals, we can directly work with actions, as each posterior belief about the state is associated with a particular selected action.⁶

Thus, the agent chooses an information-processing strategy $P : \Omega \mapsto \Delta(A)$, where $P(a|\omega)$ denotes the probability of choosing action $a \in A$ in state $\omega \in \Omega$. The unconditional probability of choosing an action a is given by:

$$P(a) = \sum_{\omega \in \Omega} P(a|\omega)\mu(\omega).$$

The corresponding posterior belief after action a is denoted by $\gamma(\cdot | a) \in \Delta(\Omega)$, where $\gamma(\omega|a)$ is the posterior probability of state ω conditional on the choice of action a . We require that all feasible attention strategies satisfy Bayes's rule so that

$$\forall \omega \in \Omega : \sum_{a \in A} \gamma(\omega|a)P(a) = \mu(\omega). \quad (1)$$

We define a learning cost function κ as

$$\kappa(P, \gamma) = \sum_{a \in A} \sum_{\omega \in \Omega} P(a)(\gamma(\omega|a) - \mu(\omega))^2. \quad (2)$$

We use a quadratic information cost function.⁷ It falls into the widely used class of posterior separable cost functions; therefore, it is linear in the induced distribution of posterior beliefs.⁸

Choosing not to learn is always an option. In this case, the chosen posterior is equal to the prior with probability one. Abusing notation, we write the problem in terms of $P(R)$, with

⁶For details, see Caplin and Dean (2013).

⁷This information cost function is also used in Wei (2021); Lipnowski et al. (2022a); Jain and Whitmeyer (2020) among others. The quadratic cost function provides us with the model tractability, allowing us to obtain a closed-form solution. In Appendix H, we present a numerical example and show that the results with the attention cost modeled as the expected reduction in entropy (Shannon, 1948; Cover and Thomas, 2012), which is most often considered in the literature, are similar.

⁸For the discussion of the posterior separable cost functions and their decision theoretic foundations see Hébert and Woodford (2021); Morris and Strack (2019); Caplin et al. (2022).

the associated action-contingent posteriors $\gamma(\cdot|R)$ and $\gamma(\cdot|S)$ understood to be chosen jointly subject to Bayes plausibility.

The agent's problem is given by

$$\max_{P(R)} \left\{ P(R) \sum_{\omega \in \Omega} v(R|\omega) \gamma(\omega|R) + (1 - P(R))S - \frac{\lambda}{2} \kappa(P, \gamma) \right\}, \quad (3)$$

subject to (1) and

$$\forall \omega \in \Omega, \forall a \in A : 0 \leq \gamma(\omega|a) \leq 1, \quad (4)$$

where $P(R)$ is the unconditional probability of choosing the risky option, $P(S) = 1 - P(R)$ is the unconditional probability of choosing the safe option, and $\frac{\lambda}{2} > 0$ is a given unit cost of information.

3.3 The option design problem

The safe option provides the agent with the certainty of receiving $S > 0$ regardless of the state of the world. The principal is purely acceptance-motivated and wants the agent to choose the risky option. The principal takes into account the agent's decision problem and decides how many utils the option will deliver in each state of the world, subject to a budget constraint $\sum_{\omega \in \Omega} v(R|\omega) \leq B$. The budget B represents the principal's capacity to generate value.

For ease of exposition and without loss of generality, we make the following assumption.⁹

Assumption 1. *The principal always uses the whole available budget:*

$$\sum_{\omega \in \Omega} v(R|\omega) = B.$$

Furthermore, we simplify the analysis by parameterizing the principal's design choice. For $\mu \geq 0.5$, define $r \in [0, \frac{B}{2}]$ as the risk level, with:

$$v(R|\omega_L) = \frac{B}{2} - r, \quad v(R|\omega_H) = \frac{B}{2} + r.$$

If $r = 0$, the principal offers a certainty-providing option (similar to the safe option). Higher values of r correspond to riskier options that concentrate payoffs in the more likely state ω_H . We refer to $r = \frac{B}{2}$ as the maximally risky option. The payoffs are summarized in Table 1.

Option/State	ω_L	ω_H
Safe (S)	S	S
Risky (R)	$\frac{B}{2} - r$	$\frac{B}{2} + r$

Table 1: Payoffs from the safe option and risky option.

⁹In Appendix E, we show that, in equilibrium, the probability of acceptance weakly increases with the available budget and, therefore, the principal would always use the entire available budget.

The following assumption ensures that the risky option cannot trivially dominate the safe option:

Assumption 2. *The budget satisfies*

$$S < B < 2S. \quad (5)$$

Thus, the principal cannot offer an option that is uniformly better than the safe option, but she can offer an option that is better than the safe option in one of the states.

Finally, the principal's problem is

$$\max_{r \in [0, \frac{B}{2}]} P^*(R), \quad (6)$$

subject to (5).

3.4 Timing and equilibrium

The timing of the game proceeds as follows:

1. The principal observes μ and selects risk level $r \in [0, B/2]$.
2. The agent observes μ and r and chooses an information acquisition strategy P .
3. Nature draws the state realization ω and signal according to P given ω .
4. The agent receives the signal, updates her prior belief to a posterior belief γ , and makes a choice a .
5. Payoffs are realized.

3.5 Equilibrium

We focus on the principal-preferred subgame perfect equilibria of this game.

Definition 1 (Equilibrium). *A subgame perfect equilibrium of the game is characterized by a pair (r^*, P^*) such that:*

1. *For any risk level r chosen by the principal, P^* constitutes a solution to (3);*
2. *r^* solves the principal's problem (6) given the agent's attention strategy P^* .*

4 Analysis and main result

We proceed by backward induction: first, we solve the agent's problem, given the principal's choice r , and then move back to the principal's problem. The solution to the agent's problem is standard and is characterized in Appendix A.

Lemma 1 characterizes the optimal option design of the principal who is ex-ante aware of how the agent decides to acquire information given the option. Figure 1 illustrates Lemma 1 for given parameters.

Lemma 1. *The principal's optimal risk level r^* is:*

a) If $\mu \in [\frac{1}{2}, \hat{\mu}]$:

$$r^* = \frac{B}{2}.$$

b) If $\mu \in (\hat{\mu}, \bar{\mu})$:

$$r^* = \frac{2S - B}{2\mu - 1}.$$

c) If $\mu \in [\bar{\mu}, 1)$:

$$r^* \in [T_1, T_2].$$

Case (a) covers all $\mu \in [0.5, 1)$ if $S \geq \frac{3}{4}B$. Case (b) is non-empty if $S < \frac{3}{4}B$. Case (c) is non-empty if $2S - B < \frac{\lambda}{2}$.

Proof. We provide the proof of the lemma and specify the formulas for $\bar{\mu}$, $\hat{\mu}$, T_1 , and T_2 in Appendix C. $\square$

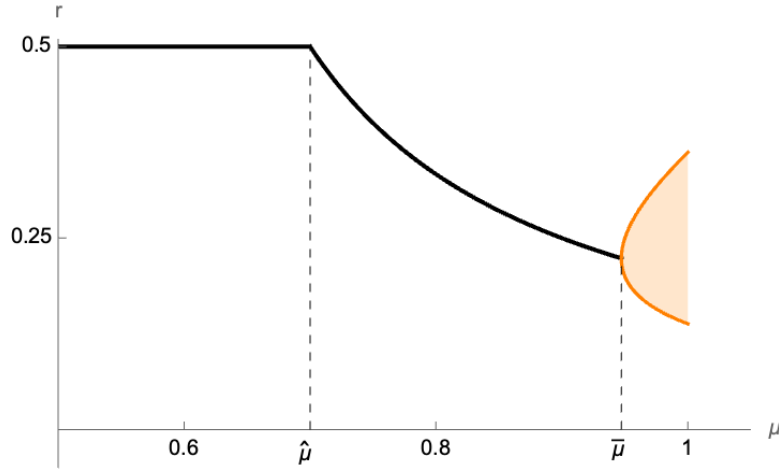


Figure 1: The principal's optimal risk level as a function of μ for $\lambda = 0.5$ and $B = 1$, $S = 0.6$. The orange area depicts the set of optimal r values that dissuade the agent from acquiring any information.

Lemma 1 distinguishes between three regions depending on μ . In case (a), when uncertainty is high (μ close to 0.5), the principal offers the maximally risky option. In case (b), when uncertainty is intermediate, the principal moderates risk according to the interior solution. In case (c), when uncertainty is low (μ close to 1), the principal can propose multiple options that dissuade the agent from acquiring information and lead her to accept blindly (see Corollary 1). The existence of these regions depends on parameters: case (b) is non-empty only when $S < \frac{3}{4}B$, while case (c) is non-empty only when $2S - B < \frac{\lambda}{2}$. When $S \geq \frac{3}{4}B$, the principal always offers the maximally risky option for all μ .

Corollary 1 (Non-learning regions). *If $2S - B < \frac{\lambda}{2}$, then for prior beliefs $\mu \in [\bar{\mu}, 1)$, the agent does not acquire information given the principal's optimal option design.*

Note that when $(2S - B)$ exceeds the marginal cost of information $\frac{\lambda}{2}$, we have $\bar{\mu} = 1$. Consequently, case (c) is empty and no non-learning region arises. Similarly, when $S \geq \frac{3}{4}B$, we have $\hat{\mu} \geq 1$, so case (b) is empty and $r^* = \frac{B}{2}$ for all μ . When both cases (b) and (c) are non-empty, the agent acquires information in cases (a) and (b), and there is a unique solution to the principal's problem in these regions.

When the principal's optimal design is not unique, for the ease of exposition, we focus on a particular optimal selection that extends the interior solution in case (b) continuously into the multiple-solution region.

Remark 1 (Selection convention). *In the non-learning region characterized in Lemma 1(c), the principal is indifferent among all risk levels $r \in [T_1, T_2]$, since each induces blind acceptance by the agent. Throughout the paper, we select*

$$r^*(\mu) = \frac{2S - B}{2\mu - 1},$$

which coincides with the unique optimal design in Lemma 1(b) and extends it continuously into the multiple-solution region. Acceptance outcomes in this region are invariant to the selection, whereas welfare comparisons depend on it.

As Lemma 1 shows, for μ close to 0.5 (high uncertainty), the principal proposes the maximally risky option $r^* = B/2$. However, as μ increases (the environment becomes more certain), the principal moderates the risk level. This result is formalized in the following proposition:

Proposition 1 (Risk Moderation). *An optimal risk level r^* weakly decreases in μ for $\mu \in [0.5, 1)$.*

Proof. See Appendix D. □

The intuition is as follows. When uncertainty is high ($\mu \in [0.5, \hat{\mu}]$), to compete with the safe option, the principal must offer a maximally risky option that provides a high payoff in the most likely state. Thus, the principal maximizes the stakes between states to incentivize the agent to acquire as much information as possible. As certainty increases, the principal can exploit the agent's reduced incentives to acquire information by offering a more moderate option that discourages learning while still being attractive enough to secure acceptance.

5 Implications for the agent's welfare

This section analyzes the consequences of strategic option design on the agent's welfare. We first establish that a benevolent principal, who maximizes the agent's expected utility, would always choose a maximally risky option. We then show that the strategic principal's design creates welfare losses that, counterintuitively, grow as the environment becomes more certain.

5.1 The benevolent benchmark

Consider a benevolent principal who maximizes the agent's expected utility rather than acceptance probability.

Proposition 2. *The benevolent principal proposes the maximally risky option: $r = \frac{B}{2}$ for all $\mu \in [0.5, 1)$.*

Proof. See Appendix F. □

The intuition is as follows. When the risky option pays B in state ω_H and 0 in state ω_L , the agent can acquire information and accept the risky option only when she learns the state is likely ω_H . If the state is likely ω_L , she takes the safe option S . This selective acceptance generates higher expected utility than any less risky design would provide. To see this, note that with a maximally risky option, when the agent can perfectly distinguish states, her expected utility (not including information costs) is:

$$\mu \cdot B + (1 - \mu) \cdot S.$$

With a less risky option that pays $\frac{B}{2} + r$ in ω_H and $\frac{B}{2} - r$ in ω_L for $r < \frac{B}{2}$, even perfect state discrimination yields lower expected utility.

This result establishes an important benchmark: any departure from maximal risk by the strategic principal represents a distortion that reduces the agent's welfare relative to the efficient outcome.

5.2 Welfare non-monotonicity

We now turn to our main welfare result. How does the agent's equilibrium welfare vary with the prior probability μ ? One might expect welfare to increase with μ . After all, an increase in μ both raises the likelihood of the favorable state ω_H and makes the environment more predictable. Therefore, in a standard decision problem without strategic design, greater certainty weakly benefits the decision-maker by reducing the value of costly information acquisition while improving expected outcomes.

However, when the principal strategically designs the option, this intuition can fail. Greater certainty enables the principal to better exploit the agent's attention constraints, and this strategic effect can dominate the direct benefit of facing a more predictable environment.

Proposition 3 (Welfare Non-Monotonicity). *Let*

$$r^*(\mu) = \begin{cases} \frac{B}{2}, & \text{if } \mu \in [\frac{1}{2}, \hat{\mu}] , \\ \frac{2S-B}{2\mu-1}, & \text{if } \mu \in (\hat{\mu}, 1). \end{cases}$$

Then agent welfare $U(\mu)$ satisfies:

$$a) \text{ For } \mu \in [\frac{1}{2}, \hat{\mu}], \frac{dU}{d\mu} \geq 0.$$

b) For $\mu \in (\hat{\mu}, \bar{\mu})$, $\frac{dU}{d\mu} < 0$.

c) For $\mu \in [\bar{\mu}, 1)$, $\frac{dU}{d\mu} = 0$.

If $S \geq \frac{3}{4}B$, case (b) is empty and welfare is weakly increasing for all $\mu \in [\frac{1}{2}, 1)$.

Proof. See Appendix G. □

When uncertainty is high (μ close to $\frac{1}{2}$), the principal offers the maximally risky option $r^* = \frac{B}{2}$. This coincides with what a benevolent principal would offer. In this region, increasing μ has only the direct effect: the agent faces a more favorable distribution over states (the high-payoff state ω_H is more likely), and her welfare increases. There is no strategic distortion because the principal cannot do better than maximal risk - any moderation would reduce acceptance probability. The condition $S < \frac{3}{4}B$ determines whether the interior region (case b) is non-empty. This condition requires that the safe option not be too attractive relative to the principal's budget. When $S \geq \frac{3}{4}B$, the safe option is sufficiently valuable that the principal must always offer maximum risk to compete: there is no room for strategic moderation, and the welfare increases monotonically in μ .

When uncertainty is low ($\mu > \hat{\mu}$), the principal moderates the option to $r^* = \frac{2S-B}{2\mu-1} < \frac{B}{2}$. Now an increase in μ has two effects:

1. **Direct effect (positive):** Higher μ means state ω_H is more likely, which benefits the agent holding the option design fixed.
2. **Strategic effect (negative):** Higher μ causes the principal to further moderate the option (by Proposition 1, r^* decreases in μ). This moderation reduces the stakes, discourages information acquisition, and lowers the agent's option value.

The key insight is that, in the interior region, the strategic effect dominates the direct effect. As μ increases, the principal can propose an increasingly "safe" option that still exceeds the outside option in expectation. The agent, facing lower stakes, rationally acquires less information and accepts the moderate option without fully investigating. The welfare loss from accepting a suboptimal option outweighs the benefit of facing a more certain state distribution.

The characterization in Proposition 3 is stated under the selection convention in Remark 1. In the non-learning region, the principal is indifferent among all $r \in [T_1, T_2]$, and alternative optimal selections within this interval can imply welfare that rises or falls with μ . Under our selected continuation, welfare is constant because $r^*(\mu) = \frac{2S-B}{2\mu-1}$ implies $U(\mu) = 2S - \frac{B}{2}$.

Figure 2 illustrates the welfare comparison between the strategic principal (black line) and the benevolent principal (blue line).

The welfare non-monotonicity result highlights a fundamental tension in the design of a risky option with a rationally inattentive agent. Greater environmental certainty, which typically benefits a decision-maker, can harm her when faced with a strategic designer who exploits her attention constraints. This suggests that in settings where principals design products or contracts for inattentive agents, efforts to reduce fundamental uncertainty may have unintended

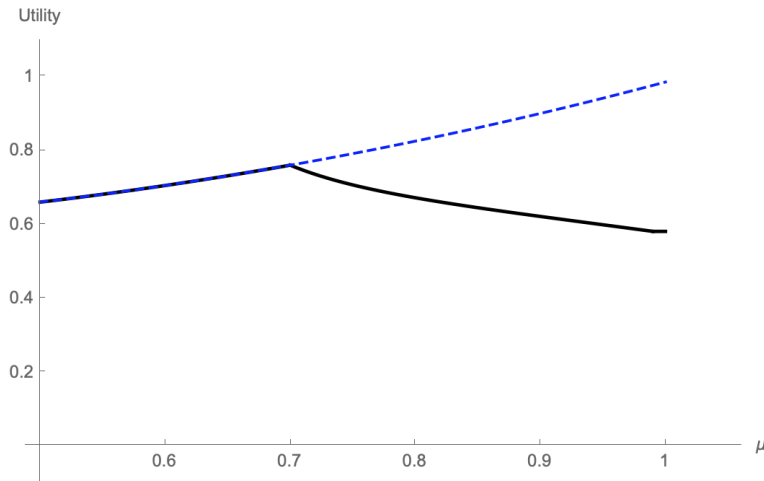


Figure 2: Agent’s expected utility under the strategic principal (black) and benevolent principal (blue) as a function of μ , for $B = 1$, $S = 0.6$, $\lambda = 0.5$. The gap between the curves represents welfare loss from strategic design.

consequences. Thus, information policies, such as mandatory disclosure or economic forecasts, could enable more effective exploitation of agent inattention, potentially harming the very agents the disclosure was intended to help.

6 Concluding remarks

The central insight of the paper is that payoff dispersion determines the value of information. When agents face costly information acquisition, compressing payoffs across states reduces the marginal return to learning and can therefore deter attention altogether. A principal who seeks acceptance rather than informed decision-making exploits this channel by moderating risk as the environment becomes more certain.

The model’s logic is independent of the particular interpretation of states, actions, or payoffs. What matters is that (i) the agent can opt out via a safe alternative, (ii) the agent observes the payoff profile before acquiring information, and (iii) the principal benefits from acceptance but not from the agent’s information. In such environments, improvements in the precision of fundamentals expand the principal’s ability to reduce stakes without jeopardizing acceptance, generating the welfare distortion.

To fix ideas, consider electoral platform design. The risky option corresponds to a proposed reform whose payoff depends on an underlying economic state, while the safe option represents the status quo. A purely office-motivated politician commits to a reform proposal before the election, when the relevant state is not yet observed. Fiscal and political constraints limit the extent to which the platform can promise high payoffs in all contingencies, so platform design involves a trade-off in state-contingent returns. The parameter governing ex-ante certainty captures voters’ prior confidence about whether the reform will succeed. Voters observe the proposed reform (the payoff profile) before deciding how much attention to devote to learning the underlying state.

When uncertainty is high, a politician must propose a high-stakes reform to induce voters to pay attention and assess its consequences. As fundamentals become more predictable, however, the politician can instead propose a moderate reform that improves on the status quo in expectation while discouraging voter scrutiny. In this interpretation, risk moderation does not reflect ideological moderation or voter preferences per se, but rather a strategic response to limited attention.

This generates a form of reform reluctance: as voters become more certain about economic conditions, the politician moderates his platform to exploit voters' inattention, even when bold reform would be welfare-improving. Therefore, a bold reform is less likely to be proposed precisely when voters are most sure it will succeed.¹⁰

The model thus highlights a general tension between informational efficiency and incentive design. By focusing on stake design rather than information control, the analysis complements existing work on persuasion and disclosure and identifies a distinct channel through which rational inattention can be exploited.

¹⁰There is a large body of literature that studies other mechanisms behind resistance to reform, for example, uncertainty regarding the distribution of gains and losses Fernandez and Rodrik (1991), pandering Canes-Wrone et al. (2001), trade-offs between welfare in the present and the future Jacobs (2016), or political culture Gratton et al. (2024).

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A Agent's optimal information acquisition

Lemma 2 characterizes the optimal probability of choosing between the two options and the posterior beliefs of the agent who takes the principal's design as given.

Lemma 2. *The agent's optimal unconditional probabilities $P^*(a)$ of choosing action $a \in \{S, R\}$ and corresponding posterior beliefs $\gamma^*(\omega_H|a)$ are:*

a) If $\gamma_1 < \mu < \gamma_2$,

$$\begin{aligned}\gamma^*(\omega_H|R) &= \max \left\{ 0, \min \left\{ 1, \frac{1}{4} \left(2 - \frac{B-2S}{r} + \frac{2r}{\lambda} \right) \right\} \right\}, \\ \gamma^*(\omega_H|S) &= \max \left\{ 0, \min \left\{ 1, \frac{1}{4} \left(2 - \frac{B-2S}{r} - \frac{2r}{\lambda} \right) \right\} \right\}, \\ P^*(R) &= \frac{\mu - \gamma^*(\omega_H|S)}{\gamma^*(\omega_H|R) - \gamma^*(\omega_H|S)}, \\ P^*(S) &= 1 - P^*(R).\end{aligned}$$

b) Otherwise,

$$\begin{aligned}\gamma^*(\omega_H|R) &= \gamma^*(\omega_H|S) = \mu, \\ P^*(R) &= \begin{cases} 1 & \text{if } \mu > \frac{S}{B}, \\ 0 & \text{if } \mu \leq \frac{S}{B}, \end{cases} \\ P^*(S) &= 1 - P^*(R).\end{aligned}$$

where $\gamma_1 = \min(\gamma^*(\omega_H|S), \gamma^*(\omega_H|R))$ and $\gamma_2 = \max(\gamma^*(\omega_H|S), \gamma^*(\omega_H|R))$.

Proof. See Appendix B. □

Lemma 2 distinguishes between two possibilities. In case (a), the agent acquires information, learns either fully or partially about the realization of the state of the world, and makes a choice based on this information. In case (b), the agent chooses based on her prior belief without acquiring additional information, because the incentives to acquire information are low compared to the cost of acquiring information.

B Proof of Lemma 2

First, we focus on the case when both options are selected with a non-zero probability, $P(a) > 0 \forall a \in A$. Because both the cost function and the expected utility are additively separable in the posteriors, we can write the objective function in terms of a net utility

$$\sum_{a \in A} P(a) \sum_{\omega \in \Omega} v(a|\omega) \gamma(\omega|a) - \frac{\lambda}{2} \sum_{a \in A} \sum_{\omega \in \Omega} P(a) (\gamma(\omega|a) - \mu(\omega))^2 = \sum_{a \in A} P(a) N(\gamma(a)),$$

where $v(a|\omega)$ corresponds to utils which the agent receives from choosing option a given state ω , and the net utility $N(\gamma(a))$ is

$$N(\gamma(a)) = \sum_{\omega \in \Omega} \gamma(\omega|a) v(a|\omega) - \frac{\lambda}{2} \sum_{\omega \in \Omega} (\gamma(\omega|a) - \mu(\omega))^2.$$

Thus, instead of maximizing the expected utility minus the cost of information for each action and corresponding posterior belief pair, we can characterize the agent's problem as a maximization of the weighted average of act-specific net utilities. As Caplin et al. (2019) show, a necessary condition for optimality is that the slope of the net utility function is the same for each chosen action at its associated posterior. We denote the posterior beliefs for the action a as $\gamma(\omega_H|a)$ and $\gamma(\omega_L|a) = 1 - \gamma(\omega_H|a)$. The slope of the net utility function is

$$\frac{\partial N(\gamma(a))}{\partial \gamma(\omega_H|a)} = v(a|\omega_H) - v(a|\omega_L) - 2\lambda(\gamma(\omega_H|a) - \mu),$$

and the same slope condition gives

$$\gamma(\omega_H|R) - \gamma(\omega_H|S) = \frac{v(R|\omega_H) - v(R|\omega_L) - v(S|\omega_H) + v(S|\omega_L)}{2\lambda} = \frac{r}{\lambda}. \quad (7)$$

Further, when both posterior beliefs $\gamma(\omega_H|a) \forall a \in A$ lie between 0 and 1, we can apply the concavification method (Kamenica and Gentzkow, 2011; Caplin and Dean, 2013) to find the posterior beliefs. Specifically, when the action space is binary, the binary attention strategy is incentive compatible if and only if the affine function connecting $(\gamma(\omega_H|S), N(\gamma(S)))$ and $(\gamma(\omega_H|R), N(\gamma(R)))$ lies above $N(\gamma(\cdot))$ on the interval $[\gamma(\omega_H|S), \gamma(\omega_H|R)]$. The tangency condition of concavification requires that

$$\frac{\partial N(\gamma(S))}{\partial \gamma(\omega_H|S)} = \frac{N(\gamma(R)) - N(\gamma(S))}{\gamma(\omega_H|R) - \gamma(\omega_H|S)}.$$

After substituting the previous results, we obtain the optimal interior posteriors that are between 0 and 1. Note that this implicitly excludes cases when, for instance, $r \approx 0$. The optimal interior posteriors are

$$\begin{aligned} \gamma(\omega_H|R) &= \frac{1}{4} \left(2 - \frac{B - 2S}{r} + \frac{2r}{\lambda} \right), \\ \gamma(\omega_H|S) &= \frac{1}{4} \left(2 - \frac{B - 2S}{r} - \frac{2r}{\lambda} \right). \end{aligned}$$

Otherwise, the posteriors are in the corner solutions. Thus, the full characterization of the

posteriors is given by

$$\begin{aligned}\gamma(\omega_H|R) &= \max \left\{ 0, \min \left\{ 1, \frac{1}{4} \left(2 - \frac{B-2S}{r} + \frac{2r}{\lambda} \right) \right\} \right\}, \\ \gamma(\omega_H|S) &= \max \left\{ 0, \min \left\{ 1, \frac{1}{4} \left(2 - \frac{B-2S}{r} - \frac{2r}{\lambda} \right) \right\} \right\}.\end{aligned}$$

So far, we focused only on the cases when the agent acquires information, i.e., if the prior belief μ is between the posterior beliefs, $\gamma(\omega_H|S) < \mu < \gamma(\omega_H|R)$. When the agent does not acquire any information, the posterior belief equals the prior belief.

Optimal unconditional probabilities $P^*(a)$, $\forall a \in A$ are obtained by using condition (1), i.e., $\forall \omega \in \Omega : \sum_{a \in A} \gamma(\omega|a)P(a) = \mu(\omega)$.

C Proof of Lemma 1

We proceed in two steps. First, we derive the optimal risk level conditional on the agent acquiring information. Second, we characterize the conditions under which the principal can induce the agent not to acquire information.

First, focus on the case when both options are selected with a non-zero probability, $P(a) > 0$, $\forall a \in A$. The principal selects the risk level to maximize the unconditional probability of acceptance. Applying condition (1), we obtain that the principal's objective function is:

$$\max_{r \in [0, \frac{B}{2}]} \frac{\mu - \gamma(\omega_H|S)}{\gamma(\omega_H|R) - \gamma(\omega_H|S)},$$

where the posterior beliefs $\gamma(\omega_H|R)$ and $\gamma(\omega_H|S)$ are functions of r . The first order condition of the objective function equals zero for

$$r = \frac{2S - B}{2\mu - 1}. \quad (8)$$

Moreover,

$$\left. \frac{d^2 P(R)}{dr^2} \right|_{r = \frac{2S-B}{2\mu-1}} = -\frac{\lambda(2S-B)}{2r^4} < 0.$$

Hence, this critical point is the unique interior maximizer.

Given Assumption 2, i.e., $B < 2S$, the numerator of formula (8) is always positive. Since $\mu \geq 0.5$, we have $2\mu - 1 \geq 0$, so $r^* \geq 0$. Note also that both candidates for the optimal $r = \left\{ \frac{B}{2}, \frac{2S-B}{2\mu-1} \right\}$ are independent of λ . λ influences the parameter space in which the agent decides not to acquire any information, but does not affect the optimal choice of option for the interior solution.

As we have shown, the objective attains a maximum at $r = \frac{2S-B}{2\mu-1}$ unless it achieves the boundary. Thus, we can characterize when $\frac{2S-B}{2\mu-1} \geq \frac{B}{2}$. It is straightforward to obtain that if $\mu \leq \hat{\mu} = \frac{2S}{B} - \frac{1}{2}$, then $\frac{2S-B}{2\mu-1} \geq \frac{B}{2}$.

To sum up, conditional on the agent acquiring information, the optimal risk level is $r^* = \frac{B}{2}$ if $\mu \leq \hat{\mu}$, and $r^* = \frac{2S-B}{2\mu-1}$ if $\mu > \hat{\mu}$. This establishes cases (a) and (b) of the lemma.

Note that $\hat{\mu} < 1$ requires $\frac{2S}{B} - \frac{1}{2} < 1$, which holds if and only if $S < \frac{3}{4}B$. When $S \geq \frac{3}{4}B$, we have $\hat{\mu} \geq 1$, so case (a) covers all $\mu \in [0.5, 1)$ and case (b) is empty.

Second, we consider when the principal can offer such an option that it is accepted with unconditional probability 1. This happens when the agent does not acquire any information and hence her posterior belief equals the prior belief and $P(R) = 1$.

According to Lemma 2, the agent does not acquire information if i) $\mu < \gamma_1$ or ii) $\mu > \gamma_2$, where $\gamma_1 = \min(\gamma^*(\omega_H|S), \gamma^*(\omega_H|R))$ and $\gamma_2 = \max(\gamma^*(\omega_H|S), \gamma^*(\omega_H|R))$.

By comparing the posteriors we get that $\gamma(\omega_H|R) > \gamma(\omega_H|S)$ if $r > 0$. We know that for $\mu > \frac{1}{2}$, only case ii) is relevant. Hence, we can consider $\mu > \gamma(\omega_H|R)$. By comparing posteriors, we get that this occurs when the prior belief is sufficiently high:

$$\mu \geq \frac{1}{2} + \frac{2S-B}{4r} + \frac{r}{2\lambda}.$$

The right-hand side of this condition depends on the risk level r . By rearranging, we get that in the non-learning region, the optimal r has to satisfy

$$2r[r - \lambda(2\mu - 1)] \leq (B - 2S)\lambda.$$

There exist multiple optimal risk levels r satisfying this condition. We solve the quadratic equation given by the previous condition. We obtain that all risk levels r that satisfy $r \in [T_1, T_2]$ are optimal and lead the agent not to acquire any information, where

$$T_1 = \max \left\{ 0, \frac{1}{4} \left(4\lambda\mu - 2\lambda - \sqrt{(4\lambda\mu - 2\lambda)^2 + 8\lambda(B - 2S)} \right) \right\},$$

$$T_2 = \min \left\{ \frac{B}{2}, \frac{1}{4} \left(4\lambda\mu - 2\lambda + \sqrt{(4\lambda\mu - 2\lambda)^2 + 8\lambda(B - 2S)} \right) \right\}.$$

This establishes case (c) of the lemma.

We can now identify the set of priors for which such r exists. By solving $T_1 = T_2 = \frac{2S-B}{2\mu-1}$ we can find the priors for which the non-learning region exists. Therefore, the agent does not acquire information for $\mu \geq \bar{\mu}$, where

$$\bar{\mu} = \frac{1}{2} + \sqrt{\frac{2S-B}{2\lambda}}$$

Note that $\bar{\mu} < 1$ requires $2S - B < \frac{\lambda}{2}$. When $2S - B \geq \frac{\lambda}{2}$, we have $\bar{\mu} \geq 1$, so case (c) is empty.

D Proof of Proposition 1

We know that when the agent acquires information, i.e., $\mu \in [0.5, \bar{\mu})$, the optimal risk level is either on the boundary and independent of μ , $r = \frac{B}{2}$, or has an interior solution, $r = \frac{2S-B}{2\mu-1}$.

Moreover, if $\mu > \bar{\mu}$, the interior solution, while not unique, is still optimal.

The first derivative of the interior solution for r is

$$\frac{\partial}{\partial \mu} \frac{2S - B}{2\mu - 1} = -\frac{2(2S - B)}{(2\mu - 1)^2} < 0.$$

Therefore, an optimal risk level r weakly decreases in μ .

E Budget utilization

We show that, in equilibrium, the probability of acceptance weakly increases with the available budget and, therefore, the principal would always use the whole budget.

First, when the agent chooses two imperfect signals, $0 < \{\gamma(\omega_H|R), \gamma(\omega_H|S)\} < 1$, we obtain that

$$\frac{\partial P(R)}{\partial B} = \frac{\lambda}{4r^2} > 0.$$

Second, when the agent receives one state-revealing signal, her optimal posteriors are $\gamma(\omega_H|R) = 1$ and $0 < \gamma(\omega_H|S) < 1$. Thus, the agent learns the state perfectly when choosing R (only chooses R in state ω_H). Taking the derivative of the acceptance probability, we obtain

$$\frac{\partial P(R)}{\partial B} = \frac{1 - \mu}{(1 - \gamma(\omega_H|S))^2} \cdot \frac{1}{4r} > 0.$$

Finally, when the agent acquires two fully state-revealing signals, the probability of acceptance is constant and equals μ .

Combining all cases, the probability of acceptance weakly increases with the available budget:

$$\frac{\partial P(R)}{\partial B} \geq 0.$$

Consequently, the principal (weakly) benefits from a higher budget and would always use the entire available budget in equilibrium.

F Proof of Proposition 2

The benevolent principal maximizes the agent's expected utility:

$$\max_{r \in [0, \frac{B}{2}]} U(r) = \sum_{a \in \{R, S\}} P(a) \left[\sum_{\omega \in \Omega} \gamma(\omega|a) v(a|\omega) - \frac{\lambda}{2} \sum_{\omega \in \Omega} (\gamma(\omega|a) - \mu(\omega))^2 \right]. \quad (9)$$

From this maximization problem, we obtain four interior candidates and two corner solu-

tions. The interior candidates are:

$$\begin{aligned} r_1 &= -\frac{\sqrt{\lambda(B-2S)}}{\sqrt{2}}, \\ r_2 &= \frac{\sqrt{\lambda(B-2S)}}{\sqrt{2}}, \\ r_3 &= \frac{\lambda(2\mu-1)}{2} - \frac{1}{2}\sqrt{\lambda(\lambda(1-2\mu)^2 - 2B + 4S)}, \\ r_4 &= \frac{\lambda(2\mu-1)}{2} + \frac{1}{2}\sqrt{\lambda(\lambda(1-2\mu)^2 - 2B + 4S)}. \end{aligned}$$

The corner solutions are $r_5 = 0$ and $r_6 = \frac{B}{2}$.

Since $B < 2S$ by Assumption 2, candidates r_1 and r_2 are undefined in real numbers. For r_3 , let $x = 2\mu - 1 \in [0, 1)$. Then $r_3 < 0$ if and only if $\lambda x < \sqrt{\lambda(\lambda x^2 + 4S - 2B)}$. Squaring both sides yields $0 < 4S - 2B$, which holds by Assumption 2. Thus $r_3 < 0$ for all $\mu \in [0.5, 1)$, and, therefore, outside the feasible domain.

For r_4 , we verify it is a local minimum by examining the second-order condition. At any critical point r , the second derivative of $U(r)$ is:

$$\frac{d^2U}{dr^2} = \frac{2}{\lambda} - \frac{2(2S-B)^2}{\lambda^2(2\mu-1)^2r^2}.$$

Evaluating at r_4 and simplifying, $\frac{d^2U}{dr^2}\big|_{r=r_4} > 0$, confirming that r_4 is a local minimum, not a maximum.

For the corner solutions: At $r = 0$, the risky option pays $B/2$ in both states. Since $B/2 < S$ by Assumption 2, the agent chooses the safe option without acquiring information, yielding $U(0) = S$. At $r = B/2$, the risky option pays B in ω_H and 0 in ω_L . The agent can acquire information and accept the risky option only when favorable, generating positive option value. Thus $U(\frac{B}{2}) \geq S = U(0)$.

Therefore, $r^* = \frac{B}{2}$. □

G Proof of Proposition 3

Case (a): For $\mu \in [\frac{1}{2}, \hat{\mu}]$, the principal offers $r^* = \frac{B}{2}$. The risky option pays 0 in state ω_L and B in state ω_H . The agent's equilibrium utility is

$$U(\mu) = S + \max \left\{ 0, \mu B - S - \frac{\lambda}{2} \kappa^*(\mu) \right\},$$

where $\kappa^*(\mu)$ is the equilibrium information cost. Since the agent can always guarantee S by ignoring the risky option, and the expected value of the risky option μB is increasing in μ , we have $\frac{dU}{d\mu} \geq 0$.

Case (b): Suppose $S < \frac{3}{4}B$, so that the interior region $(\hat{\mu}, \bar{\mu})$ is non-empty. For μ in this region, $r^* = \frac{2S-B}{2\mu-1}$. Substituting the optimal r^* and the corresponding equilibrium posteriors

and probabilities into the agent's utility, we obtain after simplification:

$$\frac{dU}{d\mu} = \frac{\lambda^2(2\mu - 1)^4 - 4(2S - B)^2}{4\lambda(2\mu - 1)^3}.$$

The numerator is strictly negative because the condition for the agent to acquire information with this r^* is exactly $\lambda(2\mu - 1)^2 < 2(2S - B)$ (see Appendix C). The denominator $4\lambda(2\mu - 1)^3 > 0$ for $\mu > \frac{1}{2}$. Therefore, $\frac{dU}{d\mu} < 0$ for all $\mu \in (\hat{\mu}, \bar{\mu})$.

If $S \geq \frac{3}{4}B$, then $\hat{\mu} \geq 1$, so case (b) is empty and case (a) applies for all $\mu \in [\frac{1}{2}, 1)$.

Case (c): Suppose case (c) of Lemma 1 is non-empty. In this region the principal is indifferent among all $r \in [T_1, T_2]$, since each such design induces blind acceptance by the agent. By Corollary 1, the agent does not acquire information in this region. Throughout the paper we therefore evaluate welfare under the selection described in Remark 1, the principal offers $r^*(\mu) = \frac{2S-B}{2\mu-1}$.

Since the agent does not acquire information and accepts the risky option unconditionally, equilibrium welfare equals the prior expected payoff from the risky option:

$$U(\mu) = \mu \left(\frac{B}{2} + r^*(\mu) \right) + (1 - \mu) \left(\frac{B}{2} - r^*(\mu) \right).$$

Simplifying and substituting the optimal r^* yields,

$$U(\mu) = \frac{B}{2} + (2\mu - 1)r^*(\mu) = 2S - \frac{B}{2}.$$

Hence welfare is constant throughout the non-learning region, and therefore

$$\frac{dU}{d\mu} = 0.$$

H Solution with the entropy cost function

We verify that our main results extend to the Shannon entropy cost function commonly used in the rational inattention literature (Shannon, 1948; Cover and Thomas, 2012). We reformulate the agent's problem as a problem of choosing conditional choice probabilities rather than posterior probabilities (Matějka and McKay, 2015).

RI agent's problem. The agent's problem is to find a vector function of conditional choice probabilities $P = \{P(a|\omega)\}_{a \in A}$ that maximizes expected payoff less information cost:

$$\max_{\{P(a|\omega)\}_{a \in A}} \left\{ \sum_{a \in A} \sum_{\omega \in \Omega} v(a|\omega) P(a|\omega) \mu(\omega) - \lambda \kappa(P) \right\}$$

subject to $P(a|\omega) \geq 0$ and $\sum_{a \in A} P(a|\omega) = 1$ for all $\omega \in \Omega$, where the unconditional choice probabilities are $P(a) = \sum_{\omega \in \Omega} P(a|\omega) \mu(\omega)$. The cost of information is $\lambda \kappa(P)$, where $\lambda > 0$ is the given unit cost of information and κ is the amount of information that the agent processes,

which is measured by the expected reduction in the entropy:

$$\kappa(P) = - \sum_{a \in A} P(a) \log P(a) + \sum_{a \in A} \sum_{\omega \in \Omega} P(a|\omega) \log P(a|\omega) \mu(\omega).$$

Using the results of Matějka and McKay (2015), the agent's optimal conditional probabilities are:

$$P(a|\omega) = \frac{P(a)e^{v(a|\omega)/\lambda}}{\sum_{a' \in A} P(a')e^{v(a'|\omega)/\lambda}},$$

where

$$P(R) = \max \left(0, \min \left(1, \frac{e^{\frac{S}{\lambda}} \left(e^{\frac{S+r}{\lambda}} - e^{\frac{B+4r}{2\lambda}}(\mu) + e^{\frac{B}{2\lambda}}(\mu-1) \right)}{-e^{\frac{B+2S}{2\lambda}} + e^{\frac{B+r}{\lambda}} + e^{\frac{2S+r}{\lambda}} - e^{\frac{B+2S+4r}{2\lambda}}} \right) \right),$$

$$P(S) = 1 - P(R).$$

The principal solves the same problem as in Equation (6). For $\mu > 0.5$, applying the same steps as in Appendix C, we obtain:

a) When the agent acquires information, the optimal risk level is:

$$r^* = \min \left\{ \frac{B}{2}, \max\{0, A\} \right\}, \quad (10)$$

where

$$A = \lambda \log \frac{\sqrt{-(e^{B/\lambda} - e^{2S/\lambda})^2(\mu-1)\mu} + e^{(B+2S)/(2\lambda)}(2\mu-1)}{e^{B/\lambda}\mu + e^{2S/\lambda}(\mu-1)}.$$

b) When the agent does not acquire information (for $\mu > 0.5$):

$$r^* \in \left[\max \left\{ 0, \lambda \log \frac{e^{-S/\lambda}(-e^{B/(2\lambda)} - Z)}{2(\mu-1)} \right\}, \min \left\{ \frac{B}{2}, \lambda \log \frac{e^{-S/\lambda}(-e^{B/(2\lambda)} + Z)}{2(\mu-1)} \right\} \right],$$

$$\text{where } Z = \sqrt{e^{B/\lambda} + 4e^{2S/\lambda}(\mu-1)\mu}.$$

Then, we use a numerical example and illustrate the solution for given parameters. Figures 3-4 illustrate the optimal risk level and the agent's utility under the entropy cost function for given parameters. The qualitative patterns mirror those in the main text with quadratic costs:

- For high uncertainty (μ near 0.5), the principal proposes maximally risky options.
- As certainty increases (μ moves toward 1), the principal moderates the risk level.
- For sufficiently high μ , the principal can dissuade the agent from acquiring any information.
- Agent's welfare is non-monotonic in μ , and, beyond some threshold, greater certainty decreases the agent's welfare.

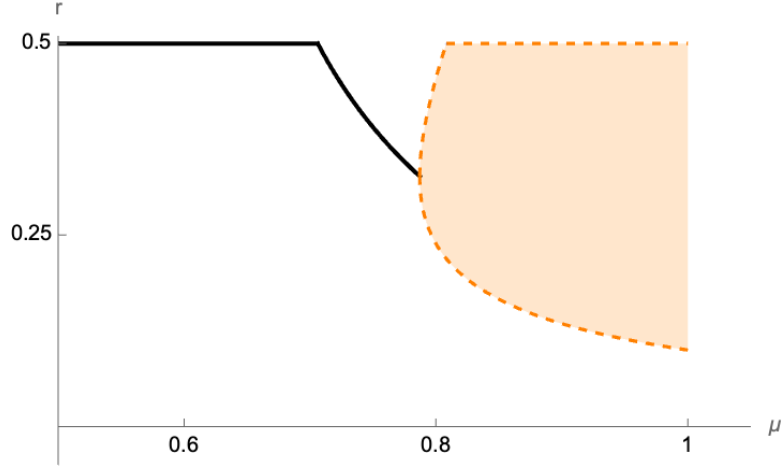


Figure 3: The principal's optimal risk level under Shannon entropy costs as a function of μ , for $B = 1$, $S = 0.6$, $\lambda = 0.5$. The orange areas depict optimal r values that dissuade the agent from acquiring any information.

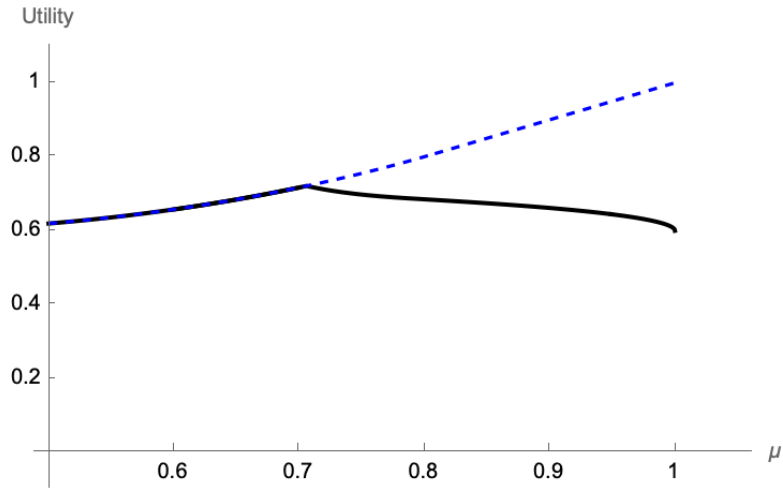


Figure 4: Agent's utility for the optimal risk level r of strategic (black line) and benevolent (blue line) principals under Shannon entropy costs as a function of μ for $B = 1$, $S = 0.6$, $\lambda = 0.5$. For the non-learning region, the interior r^* defined by equation 10 is used.